RAG Framework Comparison

LangChain vs LlamaIndex vs Haystack

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1. Introduction

Building a RAG pipeline requires selecting the right framework. LangChain, LlamaIndex, and Haystack are the three most widely adopted open-source libraries for this purpose. Each has distinct design philosophies, strengths, and trade-offs. This report compares them across key dimensions to guide framework selection for the internship RAG pipeline.

2. Framework Overview

	LangChain	LlamaIndex	Haystack
Initial Release	Oct 2022	Nov 2022	Nov 2019
Primary Focus	General LLM app framework	Data indexing & retrieval	Production search & QA pipelines
Core Abstraction	Chains & Agents	Index & Query Engine	Pipelines & Components
GitHub Stars (2025)	~90k+	~35k+	~17k+
Language	Python / JS	Python	Python
License	MIT	MIT	Apache 2.0

3. Detailed Comparison

3.1 LangChain

LangChain is a general-purpose LLM application framework with the broadest ecosystem. It excels at building complex workflows that combine retrieval, tool use, and agents.

- Pros: Extremely flexible; huge community; supports 100+ LLM providers and vector stores; rich agent & tool ecosystem; best for chaining multiple operations.
- Pros: LCEL (LangChain Expression Language) enables composable, readable pipeline construction.
- Cons: High abstraction can obscure internals; steep learning curve; over-engineered for simple RAG tasks; frequent API changes.

- Best for: Agentic RAG, multi-tool systems, complex workflows, and rapid prototyping.

3.2 LlamaIndex

LlamaIndex (formerly GPT Index) is purpose-built for connecting LLMs to data. It offers the most sophisticated data ingestion, indexing, and querying capabilities of the three.

- Pros: Best-in-class document loaders (100+ connectors); advanced chunking & indexing strategies; native support for knowledge graphs; clean query engine abstraction.
- Pros: Excellent for structured + unstructured data; strong evaluation tools built in.
- Cons: Less flexible outside of retrieval tasks; smaller agent ecosystem than LangChain.
- Best for: Data-heavy RAG, document Q&A, multi-document reasoning, knowledge base systems.

3.3 Haystack

Haystack by deepset is a production-focused NLP framework with a long history in search and question answering. It emphasizes modularity, scalability, and deployment readiness.

- Pros: Production-grade pipeline design; strong support for hybrid search (BM25 + dense); built-in evaluation framework; excellent REST API and containerization support.
- Pros: Well-suited for enterprise deployment; clear pipeline visualization.
- Cons: Smaller community than LangChain; less flexibility for agentic use cases; more boilerplate to set up.
- Best for: Production deployments, hybrid search, enterprise Q&A, teams with DevOps requirements.

4. Feature Comparison Matrix

Feature	LangChain	LlamaIndex	Haystack
Ease of Setup	Medium	Easy	Medium
RAG Support	Excellent	Excellent	Excellent
Agent Support	Excellent	Good	Basic
Data Connectors	Good (80+)	Excellent (100+)	Good (40+)
Hybrid Search	Good	Good	Excellent
Evaluation Tools	Good (LangSmith)	Excellent (built-in)	Excellent (built-in)

Feature	LangChain	LlamaIndex	Haystack
Production Readiness	Good	Good	Excellent
Community & Docs	Excellent	Good	Good
Streaming Support	Yes	Yes	Yes
Async Support	Yes	Yes	Yes

5. Recommendation

For this internship's RAG pipeline, LlamaIndex is the recommended primary framework, with LangChain as a secondary option for agent extensions.

Use Case	Recommended Framework	Reason
Basic RAG Q&A pipeline	LlamaIndex	Clean abstractions, best data ingestion, easiest to get results quickly
Agentic / multi-tool RAG	LangChain	Richest agent ecosystem and tool integrations
Production deployment	Haystack	Best REST API support, hybrid search, and DevOps tooling
Evaluation & metrics	LlamaIndex / Haystack	Both have built-in evaluation frameworks (RAGAS-compatible)

For Task 3 (Q&A pipeline notebook), LlamaIndex will be used as the primary framework due to its intuitive API, excellent document handling, and built-in support for evaluation — all critical for the subsequent tasks in this internship.

6. Conclusion

All three frameworks are capable of building production-quality RAG systems. The choice depends on the use case: LangChain for flexibility and agents, LlamaIndex for data-centric RAG, and Haystack for production search pipelines. Understanding the strengths and trade-offs of each framework is essential before implementing the RAG pipeline in Task 3.